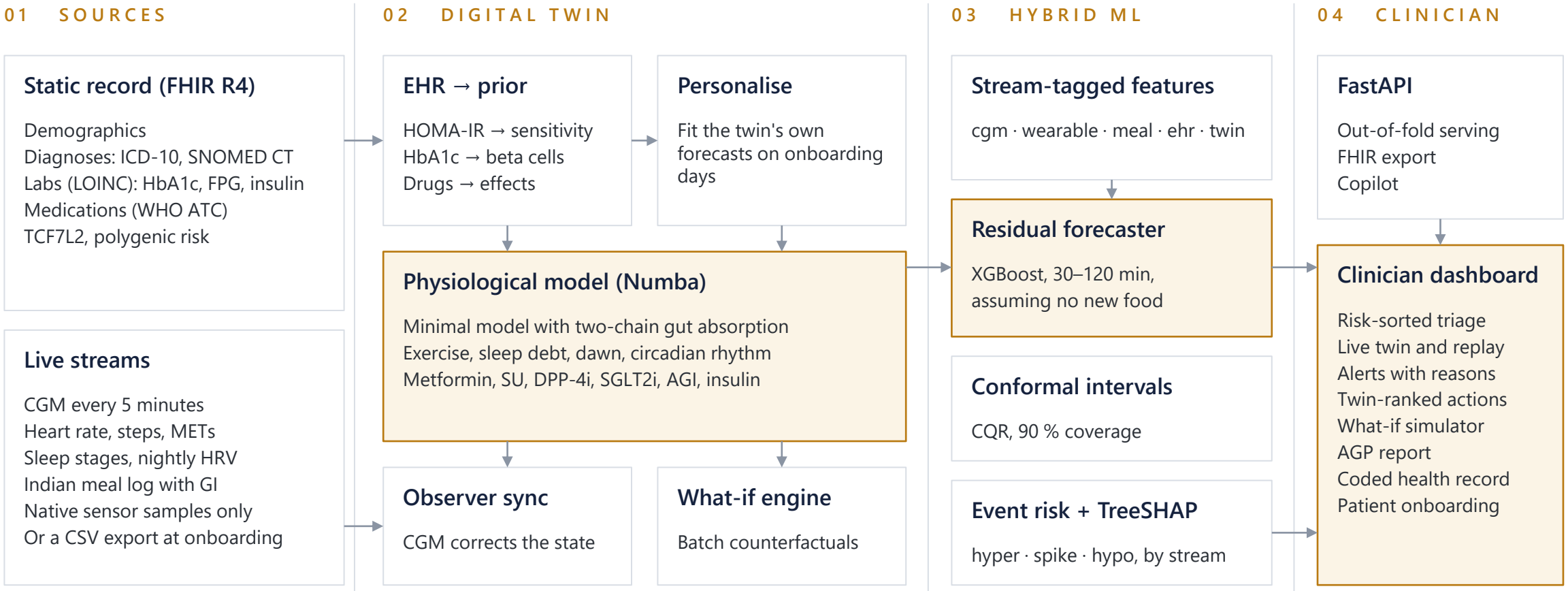
ARCHITECTURE

How MadhuTwin works: two data streams in, decisions out



GOVERNANCE Consent per patient (DPDP Act) · no dosing advice · synthetic data always labelled · out-of-fold predictions only

PIPELINE

From open data to the clinic: how every number is produced

REAL-DATA EVIDENCE · CGMACROS (PHYSIONET), 45 ADULTS

01 Download
PhysioNet open-data mirror, SHA-256 verified

→

02 Causal loader
Native sensor samples only; METs from calories

→

03 Per-patient twin
Personalised on the first 40 % (≤ 4 days); the rest is unseen

→

04 Leave-patients-out
5 folds, stratified by glycaemic group; identical rows for every model

→

05 Metrics
RMSE, Clarke grid, AUROC, lead time, coverage, meal response

DEPLOYMENT · BHARAT SYNTHETIC COHORT, 80 FICTIONAL INDIAN PATIENTS

01 Generate
Physiology-first: Indian diets, drugs, stories

→

02 Twin and features
The same pipeline as the real data

→

03 Fold models
All CGMacros plus four fifths of the cohort per fold

→

04 Out-of-fold scoring
Each patient scored by a model that never saw them

→

05 API and dashboard
Replay clock, alerts, what-if; synthetic metrics labelled

REPRODUCE `download_cgmacros → build_cgmacros → evaluate_cgmacros → build_bharat → train_deploy → uvicorn madhutwin.api.main:app`